Exploratory Data Analysis and Clustering of Earthquakes

Atmadip Goswamee ce22b045

February 2025

Objectives

1. To conduct exploratory data analysis on ISC-Gem Earthquake Dataset.
2. To explore different clustering algorithms and plot them on the world map for robust visualization.
3. Evaluate the clusters based on different metrics.

Procedure

Table 1: Snippet of the dataset (First 5 columns)

date	**lat**	**lon**	**smajax**	**strike**
1904-04-04 10:02:34.56	41.802	23.108	8.6	164.2
1904-04-04 10:26:00.88	41.758	23.249	8.3	152.2
1904-06-25 14:45:39.14	51.424	161.638	33.6	116.2
1904-06-25 21:00:38.72	52.763	160.277	28.6	43.1
1904-08-30 11:43:20.85	30.684	100.608	16.9	118.4

1. Extract the required data columns from the original `.csv` file. A snippet of the original `.csv` file is found in Table 1. The columns used to run the code are - [`'smajax', 'sminax', 'strike', 'depth', 'mo', 'lat', 'lon', 'mtt', 'mrt', 'mrr', 'mpr', 'mpp', 'mtp'`]. Any `nan` values present were dropped after this step.
2. Plot various distributions to find relationships, insights in the data.
3. Decompose the six moment tensors - [`'mtt', 'mrt', 'mrr', 'mpr', 'mpp', 'mtp'`] into their **Isotropic** and **Deviatoric** components. The **Deviatoric** component is further decomposed into two components using Eigenvector analysis.

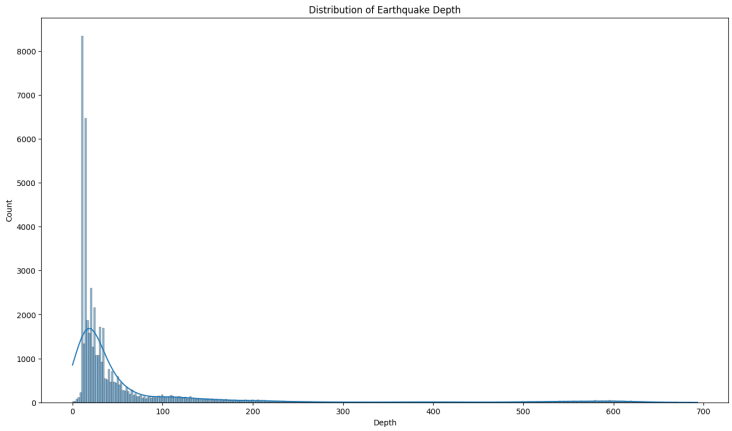

Figure 1: Histogram distribution of the depth of earthquakes.

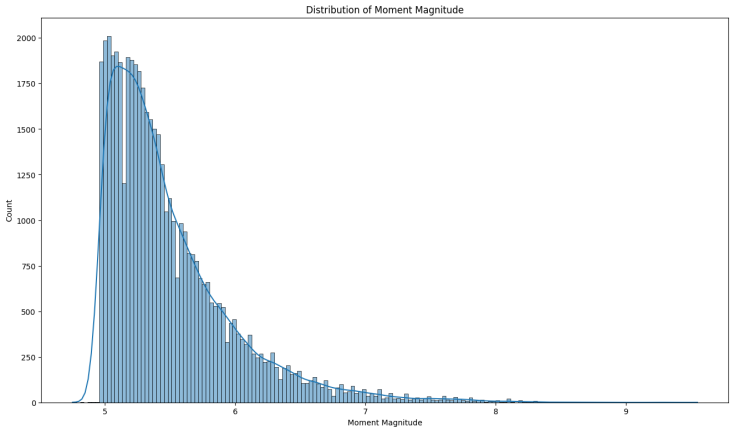

Figure 2: Histogram distribution of the Moment Magnitude of Earthquakes

(a) **Double-couple (DC)** - Represents shear faulting

(b) **Compensated linear vector dipole (CLVD)** - Represents complex source mechanisms.

The details of the implementation of the above can be found in Section 1.1

4. Use various clustering algorithms to cluster the data and visualize them on the world map. The details of the implementation can be found in Section 1.2

1 Implementation

1.1 Isotropic and Deviatropic Components

The earthquake moment tensor is constructed from six components: `mtt`, `mtp`, `mrt`, `mpp`, `mpr`, and `mrr`. For each event, the tensor is expressed as

$$M = \begin{pmatrix} m_{tt} & m_{tp} & m_{tr} \\ m_{tp} & m_{pp} & m_{pr} \\ m_{tr} & m_{pr} & m_{rr} \end{pmatrix}.$$

In our implementation, the dataset (with 47,890 events) is reshaped so that each row corresponds to a 3×3 moment tensor.

The isotropic component is computed by taking the trace of M and dividing by 3:

$$\text{iso} = \frac{\text{tr}(M)}{3}.$$

This scalar is then used to construct the isotropic matrix,

$$M_{\text{iso}} = \text{iso} \cdot I_3,$$

where I_3 is the 3×3 identity matrix. This matrix captures the volumetric (uniform) part of the seismic source.

The deviatoric component, which encapsulates the shear or distortion effects, is obtained by subtracting the isotropic part from the total moment tensor:

$$M_{\text{dev}} = M - M_{\text{iso}}.$$

To further analyze M_{dev}, an eigenvalue decomposition is performed to extract its eigenvalues λ_1, λ_2, and λ_3. These eigenvalues—sorted by their absolute values—are then used to quantify the earthquake source characteristics. In particular, the double-couple (DC) percentage is computed as:

$$\text{DC\%} = \left(\frac{|\lambda_1| - |\lambda_2|}{\max(|\lambda_1|, |\lambda_2|, |\lambda_3|)} \right) \times 100,$$

with the compensated linear vector dipole (CLVD) percentage given by:

$$\text{CLVD\%} = 100 - \text{DC\%}.$$

Finally, the scalar moment, a measure of the earthquake's energy release, is calculated via

$$M_0 = \sqrt{\frac{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}{2}},$$

which is directly related to the moment magnitude.

The vectorized computation is then completed by storing the isotropic component, DC percentage, and CLVD percentage back into the DataFrame for all events, enabling efficient and comprehensive analysis of the seismic source mechanisms.

1.2 KMeans Algorithm

The KMeans algorithm is a widely used clustering technique that partitions a dataset into k clusters by minimizing the within-cluster sum of squares. It operates by iteratively refining the cluster centroids and reassigning data points based on the Euclidean distance. The main steps of the algorithm are as follows:

1. Initialize k centroids randomly.
2. Assign each data point to its nearest centroid.
3. Recompute the centroids as the mean of all points assigned to each cluster.
4. Repeat the assignment and update steps until convergence or a predefined number of iterations is reached.

In our analysis, the optimal number of clusters was first determined using the elbow method. The elbow plot shown in Figure 3 depicts the decrease in the within-cluster sum of squares (WCSS) as the number of clusters increases. The "elbow" point in the plot suggests a good trade-off between model complexity and clustering error.

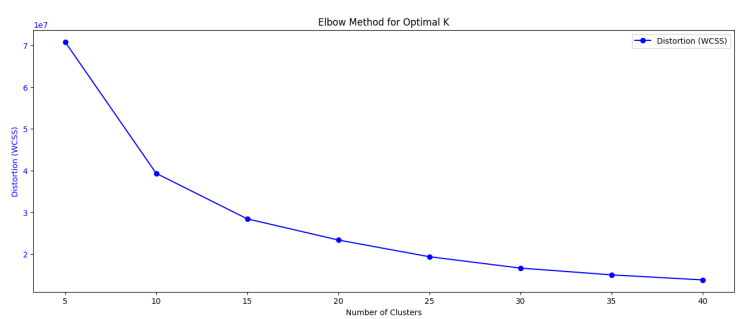

Figure 3: Elbow Plot: WCSS versus number of clusters k.

Following the selection of an optimal $k = 10$, the final clustering outcome is visualized in Figure 4. In this plot, data points are color-coded by their cluster

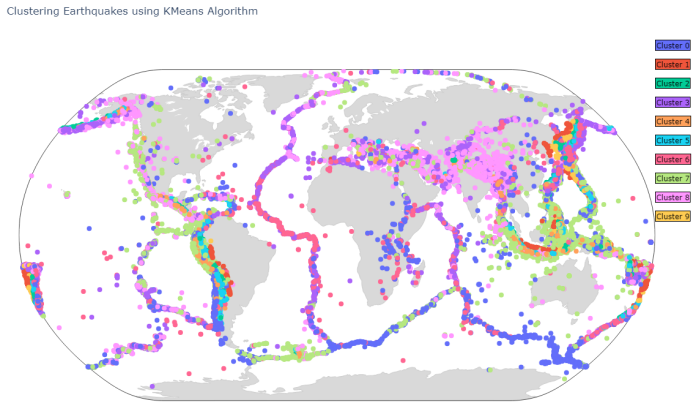

Figure 4: Cluster Visualization for the selected k using KMeans.

assignment and the computed centroids are highlighted, providing a clear view of the cluster separations.

In addition to the standard clustering visualizations, we evaluated the clustering performance using both the Silhouette score and the Davies-Bouldin index. These metrics were plotted together in Figure 5 to provide a comprehensive assessment of the clusters' cohesiveness and separation. The Silhouette score measures how similar a data point is to its own cluster compared to other clusters (with higher values indicating better-defined clusters), while the Davies-Bouldin index evaluates the average similarity between clusters (with lower values denoting better clustering quality).

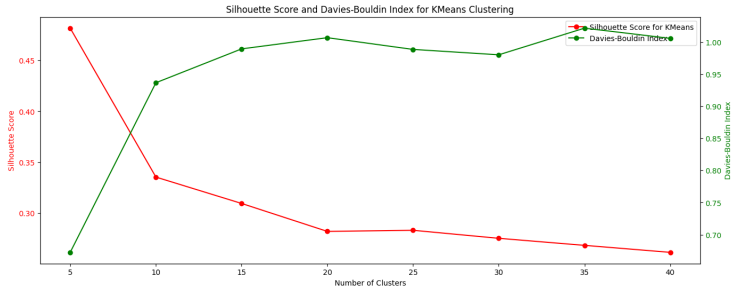

Figure 5: Combined Silhouette and Davies-Bouldin Index Plot for varying values of k.

Overall, the KMeans algorithm, combined with thorough evaluation using both the elbow method and performance metrics, provides an effective and interpretable approach for clustering the dataset. The quantitative and qualitative

assessments ensure that the final clustering structure is robust, well-separated, and meaningful.

1.3 DBSCAN

The DBSCAN algorithm operates through density connectivity analysis, formalized as:

$$\text{Core Point} \iff |N_\epsilon(p)| \geq min_samples \tag{1}$$

where $N_\epsilon(p)$ denotes points within ϵ-neighborhood of point p. Reachable points form clusters while unconnected points are classified as noise.

Parameter Optimization Framework

- **Minimum Samples:** Derived from feature space dimensionality:

$$min_samples = 2 \times d = 16 \quad (d = 8 \text{ features}) \tag{2}$$

- ϵ **Selection:** Optimized through Silhouette score maximization:

$$S = \frac{1}{N} \sum_{i=1}^{N} \frac{b_i - a_i}{\max(a_i, b_i)} \tag{3}$$

  where a_i = average intra-cluster distance, b_i = nearest-cluster distance. Grid search ($\epsilon \in [1, 25]$) yielded optimal $\epsilon = 16$ (Figure 6).

Final parameters yielded the following cluster characteristics -

- Best Parameters: $\epsilon = 16$, `min_samples` = 16
- Best Silhouette Score: 0.6523
- Clusters: 4 (with ϵ=16 verification)
- Noise Points: 763

Geospatial Validation

K-distance analysis (k=5) confirmed ϵ selection through elbow detection:

$$\text{Elbow Threshold} = argmax_\epsilon \left(\frac{d^2 y}{d\epsilon^2} \right) \tag{4}$$

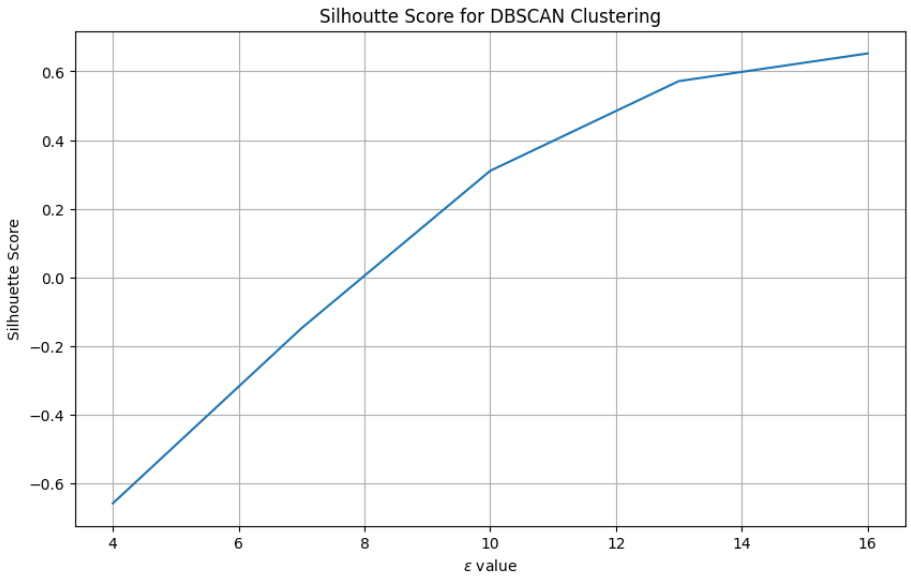

Figure 6: Silhouette scores across ϵ values with maximum at 16

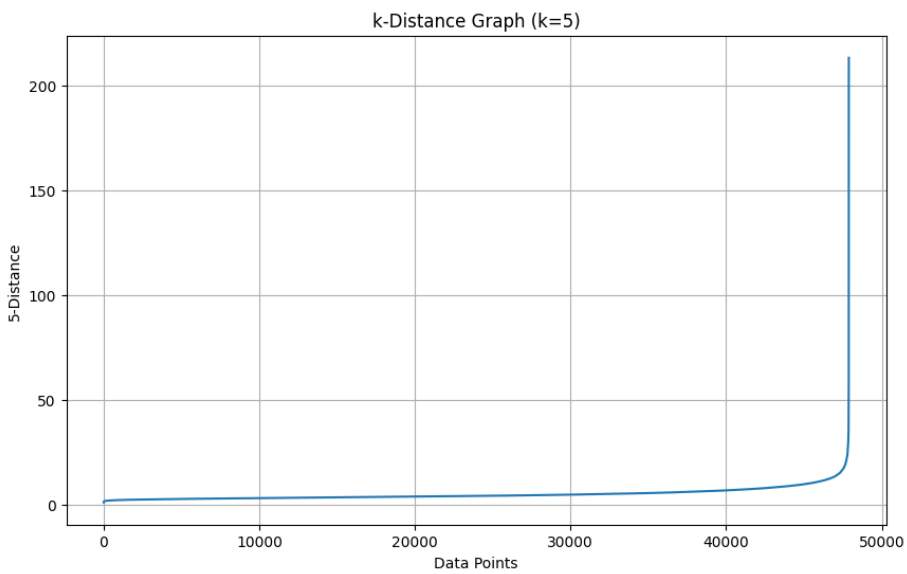

Figure 7: K-distance plot with marked elbow point at $\epsilon \approx 15$

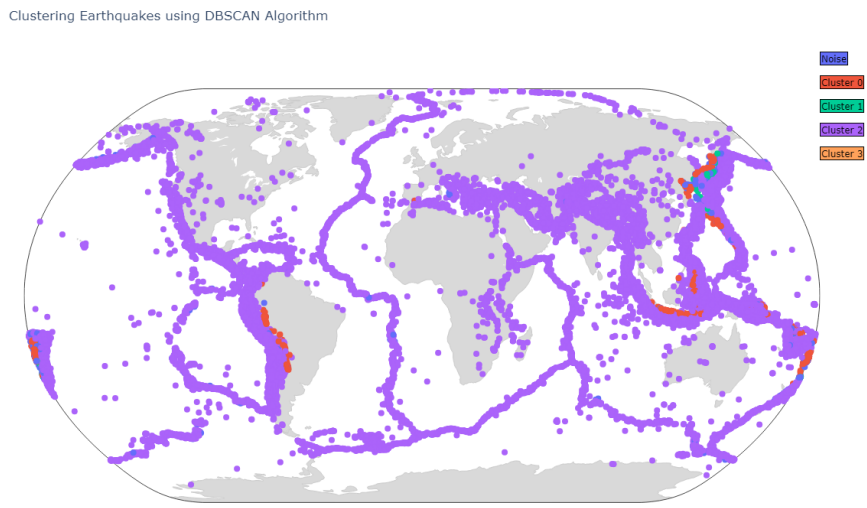

Figure 8: Earthquake clusters (colored) and noise points (blue) mapped to tectonic plates